

RAJINDER KAUR

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Patiala, Punjab - 147001, INDIA

OBJECTIVE

AI Engineer with expertise in **Large Language Models (LLMs)**, **Generative AI**, **Agentic AI**, **Retrieval-Augmented Generation (RAG)**, **Machine Learning**, and **Production AI Systems**. Independently designed, developed, and deployed **MediAgent**, a production-oriented **AI-powered medical assistant** leveraging **multi-agent orchestration**, **multimodal AI**, **vector search**, and **clinical decision support**. Experienced in building scalable end-to-end AI applications using **Python**, **FastAPI**, **LangGraph**, **PostgreSQL**, **ChromaDB**, **Redis**, **Docker**, **CI/CD**, and **AWS**. Backed by **3 years** of experience teaching and mentoring **Artificial Intelligence**, **Machine Learning**, and **Software Engineering**, with a passion for developing reliable, scalable, and production-grade AI solutions for real-world healthcare applications.

AI ENGINEERING PROJECTS

- **MediAgent – AI-Powered Medical Assistant with Multi-Agent Orchestration** *Personal Project*
[Live Demo](#)
FastAPI · React · PostgreSQL · ChromaDB · Redis · LangGraph · Groq Llama 3.3 · HuggingFace
 - Independently designed, developed, and deployed an AI-powered medical assistant leveraging **LLMs**, **Agentic AI**, **Retrieval-Augmented Generation (RAG)**, and **multimodal AI** to automate clinical workflows.
 - Architected a **LangGraph-based multi-agent system** for patient search, SOAP note generation, prescription management, appointment scheduling, clinical decision support, and medical image analysis through coordinated AI agents.
 - Built a **RAG pipeline** using **ChromaDB**, vector embeddings, semantic search, and metadata filtering to retrieve relevant patient records and generate context-aware responses.
 - Integrated **Groq Llama 3.3** with tool calling, structured outputs, prompt engineering, and intent-based agent routing to execute multi-step clinical tasks.
 - Developed **multimodal AI workflows** for automated interpretation of Chest X-rays, CT scans, MRI, ECG, and retinal images using Hugging Face vision models.
 - Built an automated **LLM evaluation pipeline** using **RAGAS** to measure Answer Relevancy, Faithfulness, Context Precision, and Context Recall, generating reproducible evaluation reports and achieving an **Answer Relevancy score of 0.6679**.
 - Engineered secure backend services using **FastAPI**, **PostgreSQL**, **Redis**, **JWT authentication**, and **RBAC**, exposing REST APIs for clinical data management and AI agent orchestration.
 - Containerized the application with **Docker** and automated testing and deployment using **GitHub Actions**; optimized inference and retrieval performance through caching and evaluation-driven improvements, achieving **1.8 s API latency**, **35 ms database queries**, **65 ms vector search latency**, and **85% automated test coverage**.

SKILLS

Generative AI & AI Engineering: Large Language Models (LLMs) • Generative AI • Agentic AI • AI Agents • Retrieval-Augmented Generation (RAG) • LangGraph • Prompt Engineering • Tool Calling • Structured Outputs • LLM Evaluation • RAG Evaluation • Embeddings • Semantic Search • Hybrid Retrieval • BM25 • Cross-Encoder Reranking • Vector Search • ChromaDB • Vector Databases

Machine Learning: Machine Learning • Natural Language Processing (NLP) • Data Preprocessing • Feature Engineering • Model Evaluation • Hyperparameter Tuning • Hallucination Detection

AI Evaluation & MLOps: MLflow • Experiment Tracking • RAGAS • DeepEval • Model Evaluation • Production Monitoring

Backend Engineering: Python • FastAPI • SQLAlchemy • REST APIs • JWT Authentication • Role-Based Access Control (RBAC) • Async Programming • API Integration

Databases: PostgreSQL • ChromaDB • Redis • SQLite • Neon

Deployment & DevOps: Docker • Docker Compose • Kubernetes • Git • GitHub Actions • CI/CD • Render • Vercel • Cloud Deployment

Software Engineering: System Design • Scalable API Development • Production AI Applications • Software Architecture • API Security

EDUCATION

Punjabi University

M.Tech. Computer Engineering

• GPA: 9.00/10.00

July 2021 - June 2023

Patiala, India

Punjabi University

B.Tech. Computer Science and Engineering

• Division: First with Distinction.

July 2017 - June 2021

Patiala, India

PROFESSIONAL EXPERIENCE

Assistant Professor - Dept. of Emerging Technologies
Chandigarh Group of Colleges (CGC)

June 2025 - Feb 2026
Landran, India

- Designed and delivered **AI and Machine Learning** learning modules using real-world datasets and practical applications.
- Mentored students in developing **AI/ML projects**, emphasizing problem-solving, system design, and structured reasoning.
- Evaluated technical solutions for **correctness, clarity, and reasoning quality**, providing actionable feedback and guidance.

Assistant Professor — Computer Science
Chandigarh University

June 2023 - June 2025
Mohali, India

- Delivered courses in **Data Structures and Algorithms, Database Management Systems, Software engineering and AI-related subjects** using structured pedagogical approaches.
- Mentored **70+ student projects**, providing guidance in system design, debugging, and software development across diverse technical domains.
- Designed and conducted interactive sessions, assessments, and feedback mechanisms to improve problem-solving and technical understanding.
- Specialized in simplifying complex systems into logically sequenced explanations for learners at different levels.
- Conducted mock interviews and problem-solving sessions focusing on reasoning, clarity, and structured thinking.

PATENTS AND PUBLICATIONS

- [1] **An Overview of Advanced Technologies Applied to Identified Printed and Handwritten Text in Gurmukhi Script: A Review.** *Journal Punjab Academy of Sciences*, Vol. 23, pp. 194–207. ISSN: 2229-7014 (Print).
- [2] **CodeXpress: Realtime Collaborative Environment** . In *2025 8th International Conference on Circuit, Power and Computing Technologies (ICCPCT)*, IEEE. DOI:10.1109/ICCPCT65132.2025.11176533
- [3] **Enhancing Trust in Healthcare Diagnosis Through Explainable AI.** In *Intelligent IT Solutions to Promote Indigenous Innovations*, Springer. DOI: 10.1007/978-981-96-3936-6
- [4] **Improved Plant Disease Identification with CNN's GLCM and KNN Fusion..** In *International Conference on Computational Intelligence and Information*, Springer. DOI: 10.1007/978-981-96-3936-6
- [5] **Device for Mobile Application Authentication.** Indian Design Patent – Granted. Application No. 460546-001.
- [6] **Sensor Based Alarm in Vehicle Adhoc Network (VAN).** Indian Design Patent – Application No. 462963-001.

AWARDS AND RECOGNITIONS

- **Certificate of Recognition – Smart India Hackathon (SIH) 2025**
Chandigarh Group of Colleges, Landran

Nov 2025 – Present

Guided teams in building **AI-based systems** including a **multilingual NLP telemedicine platform** and a **genetic algorithm-based scheduling system**.

ADDITIONAL INFORMATION

Languages: English, Punjabi, Hindi